

Recreating the Sodium Vapor Matting Process

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(a) Foreground



(b) (Flipped) Sodium Vapor Matte



(c) Composite onto new background

ABSTRACT

We present a Sodium Vapor Matting (SVM) system built from off-the-shelf components allowing simultaneous capture of a full-color foreground subject and an accurate holdout matte. We use a beamsplitter cube and two filters to image the sodium vapor wavelength independently of the rest of the visible spectrum. We build custom sodium vapor light sources for illuminating the background, and aim synchronized digital cinema cameras into the filtered prism to record the foreground and the matte. We process several composite shots using the system, using the matte and a clean plate to subtract visible light in the subject's background and to hold out the alpha region from the new background.

CCS CONCEPTS

• **Computing methodologies** → **Computational photography; Image processing.**

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Introduction. Sodium Vapor Matting (SVM) was an influential technique for filming live-action actors in a studio and compositing them onto a new background image, such as a photograph of a location or hand-painted animation. It was practiced with a specialized camera capable of shooting onto two strips of film simultaneously,

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and used a beamsplitter and filters to record the 589nm sodium vapor wavelength onto a strip of black-and-white film and the rest of the visible spectrum onto a strip of color film. With the background behind the actors illuminated with monochromatic yellow sodium vapor light, a near-perfect matte was recorded on the black-and-white film, while an effectively full-color image was recorded on the color film – everything necessary to create an optical composite of the actors onto a new background.

SVM was developed in England [Kudar 1949] and first used by the The Rank Company for the 1956 feature film *Plain Sailing* [Brosnan 1974]. It was subsequently used on numerous Disney productions including 1964's *Mary Poppins* featuring Julie Andrews with a translucent veil and Dick van Dyke dancing with animated penguins. After 1990's *Dick Tracy* the technique fell into disuse due to difficulties in maintaining (and locating) the equipment, and improvements in chroma-key technology.

SVM achieved excellent composites since a high-quality matte was recorded directly onto its own film strip, essentially creating a four-channel RGB- α camera. Previous systems which used near-infrared (NIR) light for the matte used simpler lighting and filters, but recorded poorer-quality mattes since cinema lenses were not optimized to image NIR light sharply or at the same focal plane as visible light¹. The key insight of SVM was to move the matte lighting into the visible spectrum, where it would naturally align with the visible image, and use single-wavelength sodium vapor light which could be filtered away from the foreground image without appreciably altering its colors. Furthermore, the yellow-orange wavelength of the sodium light fell between the peak sensitivities of the red and green film records, further facilitating the optical separation. [Vlahos 1963] described an optically efficient prism which separated the wavelengths at the beamsplitter interface.

¹[Debevec et al. 2002] showed that NIR compositing is more practical in the modern era using narrow-wavelength NIR LED's and digital image registration

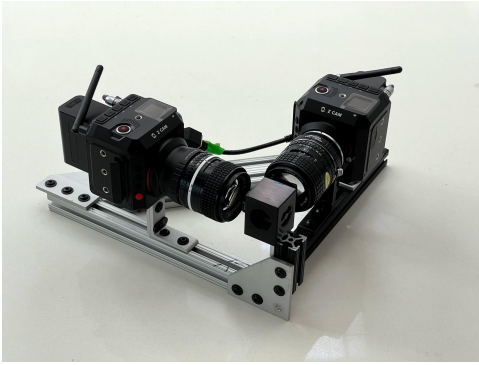


Figure 1: Dual-camera and filtered beamsplitter setup.

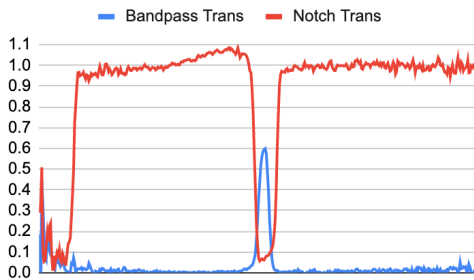


Figure 2: Filter transmission spectra.

Our System. While building an analogous SVM system with digital image sensors would be straightforward, it would need custom engineering to mount monochrome and color image sensors upon the faces of a beamsplitter cube. Our approach aims to be low cost using only off-the-shelf components. We make two design tradeoffs to achieve this: 1) use a regular beamsplitter cube and filter the light as it leaves the beamsplitter, and 2) place two complete camera-lens systems behind the filtered beamsplitter, instead of a beamsplitter behind the lens of a single multi-sensor camera. The first tradeoff halves the amount of light to each sensor, and the second requires additional image registration, but neither of these is daunting in the digital era. Fig. 1 shows our beamsplitter camera setup.

We used a Thorlabs 594nm Notch Filter (NF594-23) to remove the sodium wavelength from the color camera and an Edmund Optics 589nm Bandpass Filter (#65-707) for the matte camera, each 25mm in diameter mounted into an Edmund Optics C-Mounted Non-Polarizing Cube Beamsplitter (#49-682). Although the filter pair had slightly different center wavelengths, the transmission and rejection spectra dovetail nicely as seen in Fig. 2(a).

Our two cameras are Z-CAM E2 4K cinema cameras connected with a synchronization cable, with matched Meike 50mm MFT lenses positioned closely behind the centers of the filters on the beamsplitter using T-Slot framing.

We lit a white wall behind the subject with two Philips 321521 90-watt T15 SOX90 Low Pressure Sodium HID Light Bulbs powered through Fulham Lighting WH5-120-L Workhorse 5 Lamp Ballasts, re-housed in fluorescent light fixtures. We lit the subject with two RGB LED light wands which emit very little light at the 589nm

wavelength between the red and green LEDs, similar to the light from LED panels on a virtual production stage, which could also be a good place to practice this technique.

Recording and Processing. We shot several subjects with a wide range of foreground colors and transparency including motion blur, a veil, and a bottle of water. We also shot clean plates, a color chart, and a calibration grid of small holes for alignment. All clips were shot at 24fps with a 180 degree shutter. The footage was processed as follows:

- (1) Linearize the pixel values of the color and matte images
- (2) Flip the matte image since it was reflected in the prism
- (3) Compute the ratio matte image of the sodium matte divided by the sodium clean plate
- (4) Compute the unregistered matte as the average of the red and green channels of the ratio matte
- (5) Identify corresponding points in the calibration grid images to compute a 2D homography transformation H
- (6) Apply H to register the matte to the color image
- (7) Multiply the matte by the clean plate of the color camera to form an image of the visible spill light onto the background
- (8) Subtract the spill light from the color image so the subject is seen against a black background
- (9) Use the color chart to determine a color transformation matrix to remap the color image into a desired color space (e.g. ACES or sRGB primaries)
- (10) Multiply the new background plate by the matte image to hold out an area for the subject
- (11) Add the subject image onto the background with holdout
- (12) Tone map the linear-space final composite with a desired transfer curve (such as Rec. 709 or sRGB)

The teaser image and the accompanying video shows that our SVM technique produces high-quality composites with no matte lines and acceptable color rendition, transparency, and motion blur.

Improvements. The matte camera sensor could be monochrome without a color filter array to increase its sensitivity and effective resolution. The focus rings of the two lenses could be ganged mechanically. A 4-channel sensor with RGB and yellow pixels, ideally with minimal spectral overlap, would allow the technique to be practiced with a single camera and lens. Yellow (or amber) LEDs could be used instead of sodium vapor lamps, which are fragile and take some time to turn on. The wider spectral distribution of LEDs could require wider-bandwidth filters or additional light filtration. If practiced in an LED volume, panels with RGB and yellow LEDs could be used to create the yellow background light in the camera frustum. Accurate RGB+ α imagery from this system could be useful in training machine learning networks to solve the natural image matting problem.

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